PORTFOLIO PROJECT | DATA ANALYTICS

Financial Loan Analysis

Problem Statement & Project Scope

1. Background

A lending company issues personal loans to individual borrowers. Each loan is given a **grade (A–G)**, a **term (36 or 60 months)** and an **interest rate**, and the borrower repays in monthly instalments. Some borrowers repay in full (**Fully Paid**), some are still repaying (**Current**), and some stop paying and the loan is written off (**Charged Off**).

Management receives the monthly numbers, but there is no single view that shows **how fast the loan book is growing, who the borrowers are, which loans go bad, and which segments actually make money**. The business needs this to decide who to lend to, on what terms, and how much risk to take.

2. Problem Statement

> **How can the lender grow its loan portfolio while controlling bad loans?** The analysis must (1) monitor the portfolio through clear KPIs, (2) identify which borrower and loan characteristics are linked to charge-offs, and (3) show which segments are profitable, so that lending policy can be improved.

3. Objectives

- **Monitor performance:** build a dashboard that tracks applications, funded amount, received amount, interest rate and DTI, with month-to-date (MTD) and month-over-month (MoM) change.
- **Measure loan quality:** split the book into **good loans** (Fully Paid + Current) and **bad loans** (Charged Off) and quantify the money lost.
- **Find the drivers of risk:** compare charge-off rates by grade, term, purpose, loan amount, income, DTI, employment length, home ownership, verification status and state.
- **Find where the profit is:** compare returns by grade, purpose and term.
- **Recommend actions:** turn the findings into clear, practical lending recommendations.

4. Key Business Questions

1. How many loans were issued, how much money was lent and how much was received? How are these changing month by month?
2. What share of loans are good and bad, and how much money is lost on bad loans?
3. Which grades, terms, purposes and borrower types have the highest charge-off rates?
4. Which segments give the best and worst return on the money lent?
5. Where are the loans located, and is the portfolio too dependent on a few states?
6. What should the lender change in its approval and pricing policy?

5. Dataset

File: financial_loan.csv | **Size:** 38,576 loans × 24 columns | **Period:** January – December 2021 | **Type:** all individual applications from 50 US states.

Column group	Columns and meaning
Loan details	id, member_id (identifiers) · loan_amount (money lent) · term (36/60 months) · int_rate · installment (monthly payment) · grade, sub_grade (lender's risk rating) · purpose · issue_date
Borrower details	annual_income · dti (debt-to-income ratio) · emp_length, emp_title (employment) · home_ownership · verification_status (income verified or not) · total_acc (number of credit accounts) · address_state
Outcome details	loan_status (Fully Paid / Current / Charged Off) · total_payment (money received so far) · last_payment_date · next_payment_date · last_credit_pull_date

KPI definitions

KPI	Definition
Total Loan Applications	Number of loans (count of id)
Total Funded Amount	Sum of loan_amount – money lent out
Total Received Amount	Sum of total_payment – money collected from borrowers (principal + interest)
Average Interest Rate / DTI	Mean of int_rate / mean of dti
MTD and MoM	MTD = latest month (December 2021). MoM = % change of December compared with November
Good Loan	loan_status is Fully Paid or Current
Bad Loan	loan_status is Charged Off
Charge-off rate	Charged-off loans ÷ finished loans (Fully Paid + Charged Off). Current loans are left out because their result is not known yet
Return on money lent	(Total received – Total funded) ÷ Total funded, for finished loans

6. Approach

- **Clean & prepare (Python):** check duplicates and missing values, fix data types, handle outliers, and create new columns (good/bad flag, bands for income, DTI and loan amount, profit per loan).
- **Explore & analyse (Python – pandas, matplotlib, seaborn):** KPIs, univariate analysis, monthly trends, risk analysis, profitability and geography.
- **Visualise (Power BI):** a two-page interactive dashboard – Overview and Summary – with filters for grade, purpose and state.
- **Report:** summarise the insights with charts and give recommendations.

7. Deliverables

Deliverable	Description
Problem Statement	This document
Python Analysis (Notebook)	Financial_Loan_Analysis.ipynb – cleaning, KPIs, charts and insights
Dashboard	Page 1 Overview and Page 2 Summary (Power BI screenshots)
Insights Report	Findings with charts from the notebook, and recommendations

Tools

Python (pandas, NumPy, matplotlib, seaborn) · Power BI · Jupyter Notebook / Google Colab